

BURNS

1. WHAT IS A BURN?

A burn is damage to the skin and tissues caused by heat, chemicals, electricity, or radiation (like sunlight or X-rays).

Severity depends on:

-  Depth (how deep the burn is)
-  TBSA (how much skin is burned – Total Body Surface Area)
-  Location (face, hands, feet, genitals, airway = more serious)

2. TYPES & CLASSIFICATIONS OF BURNS

BY DEPTH (DEGREE)

Degree	Layers Affected	Symptoms	Pain?	Healing Time
1st Degree (Superficial)	Epidermis	Red, dry, blanches, no blisters	Yes	3–7 days
2nd Degree (Superficial Partial Thickness)	Epidermis + partial dermis	Blisters, pink, moist, very painful	Yes	1–3 weeks, scar risk high
2nd Degree (Deep Partial Thickness)	Epidermis + deep dermis	Pale, white, diminished pain	Less	May require graft
3rd Degree (Full Thickness)	All skin layers into fat	Leathery, dry, black/red/white, no pain	No	Months, needs graft
4th Degree (Subdermal)	Into muscle/ bone	Charred, no sensation, exposed bone	No	Amputation likely
Unstageable	All black, eschar	Nothing visible	No	Requires debridement

3. TBSA – TOTAL BODY SURFACE AREA (RULE OF NINES)

Used to estimate how much of the body is burned (TBSA) for fluid resuscitation and classification.



ADULTS

Body Area	% TBSA
Head & Neck	9%
Anterior Trunk	18%
Posterior Trunk	18%
Each Arm (Total)	9%
Each Leg (Total)	18%
Perineum/Genitals	1%



CHILDREN – ADJUSTED PERCENTAGES

Body Area	% TBSA
Head & Neck	18%
Anterior Trunk	18%
Posterior Trunk	18%
Each Arm (Total)	9%
Each Leg (Total)	14%
Perineum/Genitals	1%

4. BURN SEVERITY (MINOR, MODERATE, MAJOR)

Severity	TBSA/Location	Examples
Minor	<10% in adults, <5% in kids/elderly	Sunburn
Moderate	10–20% TBSA or deep 2nd°	Small 3rd-degree
Major	>20% TBSA or critical areas (face, hands, feet, genitals, airway)	Electrical, inhalation, large burns

5. FLUID RESUSCITATION – PARKLAND FORMULA

Used in emergent phase to prevent shock.

- Adults: $4 \text{ mL} \times \%TBSA \times \text{kg}$
- Children: $3 \text{ mL} \times \%TBSA \times \text{kg}$
- Give $\frac{1}{2}$ in first 8 hrs, then $\frac{1}{2}$ over next 16 hrs

✓ Use Lactated Ringer's

- Replaces fluids/electrolytes
- Raises pH
- Only under supervision

6. BURNS THAT AFFECT THE AIRWAY (INHALATION BURNS) - LIFE-THREATENING

 Caused by breathing in smoke, steam, or toxic gases (like carbon monoxide or cyanide).

🔥 Signs of a Dangerous Airway Burn:

- ✓ Soot (black powder) around mouth or nose.
- ✓ Singed nasal hairs (burned nose hair).
- ✓ Hoarseness (voice sounds weak or different).
- ✓ Stridor (high-pitched noise when breathing = airway is closing!).
- ✓ Confusion or Cherry-Red Skin (signs of Carbon Monoxide poisoning).

🔥🚒 IMMEDIATE ACTION: INTUBATE (insert breathing tube) BEFORE AIRWAY SWELLS SHUT! 🚒

🔑 INTUBATE EARLY before swelling closes airway!

7. BURN TREATMENT STAGES (3 PHASES OF BURN CARE)

A. EMERGENT PHASE (FIRST 24-48 HOURS) - SAVE THE LIFE!

🔥 Priority: AIRWAY, FLUIDS, PREVENT SHOCK!

- ✓ Check Airway! (Intubate if inhalation injury suspected).
- ✓ Fluid Replacement (Prevent Shock!)

Use the Parkland Formula (IV Fluids to replace lost fluids):

$4 \text{ mL} \times \text{Body Weight (kg)} \times \% \text{ TBSA burned}$

Give Half in First 8 Hours, The Rest Over Next 16 Hours

- ✓ Pain Control: Give IV opioids (morphine, fentanyl).
- ✓ Cover with Clean, Dry Sheet (Prevents infection & heat loss).

B. ACUTE PHASE (48 HOURS – WEEKS) - HEAL THE WOUND!

🔥 Priority: Wound Healing, Infection Prevention, Nutrition

- ✓ Prevent Infection (Sterile Wound Care, Antibiotic Ointments like Silver Sulfadiazine).
- ✓ Debridement (Removing Dead Tissue) - May Need Skin Grafts.
- ✓ High-Calorie, High-Protein Diet (Burns increase metabolism).

C. REHABILITATION PHASE (MONTHS – YEARS) - RESTORE FUNCTION!

🔥 Priority: Physical Therapy, Scar Prevention, Emotional Support

- ✓ Range of Motion (ROM) Exercises to Prevent Contractures (stiff, tight skin).
- ✓ Wear Compression Garments to Reduce Scars.
- ✓ Support Groups (Burn Survivors Often Have Emotional Trauma).

8. SPECIAL BURNS

ELECTRICAL BURNS

7. SPECIAL TYPES OF BURNS (ELECTRICAL & CHEMICAL BURNS)

A. ELECTRICAL BURNS (CAUSED BY LIGHTNING, POWER LINES)

-  Hidden Damage! Can injure the heart, brain, muscles, and organs!
- ✓ Monitor Heart (Risk for Deadly Arrhythmias Like Ventricular Fibrillation!).
- ✓ Check Urine for Myoglobin (Sign of Rhabdomyolysis - Muscle Breakdown That Can Damage Kidneys).
- ✓ Give High IV Fluids to Flush Kidneys!

B. CHEMICAL BURNS (CAUSED BY ACIDS, CLEANING PRODUCTS)

-  Can keep burning until fully washed off!
- ✓ Remove Contaminated Clothing.
- ✓ Flush Skin with Water for At Least 20-30 Minutes.
- ✓ If it's a dry powder chemical, BRUSH IT OFF FIRST (then rinse).

9. COMPLICATIONS TO WATCH FOR

⚠ Hypovolemic Shock (Severe Fluid Loss in First 24-48 Hours!)

Signs: Low Blood Pressure, Fast Heart Rate, Pale/Cold Skin.

Treat: Give IV Fluids (Use Parkland Formula).

⚠ Compartment Syndrome (Too Much Swelling Inside Tight Skin = Blocks Blood Flow!)

Signs: Pain, Pale Skin, No Pulse, Tingling, Paralysis.

Treat: Escharotomy (Cutting Burned Skin to Release Pressure).

⚠ Sepsis (Life-Threatening Infection in Burn Patients!)

Signs: Fever, Fast Heart Rate, Low Blood Pressure, Confusion.

Treat: IV Antibiotics, Sterile Wound Care.

10. NCLEX MUST-KNOW PRIORITIES

-  **AIRWAY FIRST** – facial burns, hoarseness, soot
-  **FLUID REPLACEMENT** – Parkland formula, LR
-  **PAIN CONTROL** – IV opioids only
-  **PREVENT INFECTION** – sterile wound care
-  **NUTRITION** – High-protein, high-calorie
-  **PSYCHOSOCIAL CARE** – long-term support

11. RAPID REVIEW

Topic	Must-Know
TBSA	Rule of Nines (adult/child)
Fluids	Parkland Formula – LR
Airway	Soot, singed hair, hoarseness → intubate
Pain	IV opioids ONLY
Electrolytes	↑K ⁺ early, ↓K ⁺ later
Infection	Fever, ↑WBC, pus, odor
Psychosocial	PTSD, anxiety, body image
NCLEX Key Terms	Moist/pink = 2nd°, Leathery = 3rd°, Bone = 4th°

12. WHY NOT IM OR PO MEDS EARLY?



In emergent burns:

-  **IM = INTRAMUSCULAR**
 - Medicine is given into the muscle using a needle
 - Absorbed into the bloodstream slowly
 - Example: flu shot

-  PO = PER OS (LATIN FOR "BY MOUTH")
 - Medicine is taken orally (swallowed)
 - •Absorbed through the digestive system

WHY NOT USE IM/PO IN BURN EMERGENCIES?

In the emergent burn phase, circulation is poor, especially to muscles and the gut.

- ➔ IM and PO meds may not absorb properly
- ➔ So IV (intravenous) meds are the safest and fastest!

 Use IV meds only